

COS 314 Assignment 3 Report

Genetic Programming for Breast Cancer Classification

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1. Introduction

- The assignment implements two Genetic Programming (GP) approaches to classify instances from the Breast Cancer Wisconsin (Diagnostic) dataset as recurrence-events (class 1) or no-recurrence-events (class 0).
 - Arithmetic Classifier (Symbolic GP): This evolves mathematical expressions using arithmetic operators (+, -, *, /) to produce a continuous score that is passed through a sigmoid function for binary classification.
 - Decision Tree Classifier (Logical GP): This evolves binary trees with feature-based splits and leaf predications.
- Both models follow standard GP principles, including population based evolution with selection, crossover and mutation but differ in representation and operators. The goal is to compare their performances on training and unseen test data, including accuracy, F-measure, runtime, and statistical significance. All code is implemented fully in Java, seeded for reproducibility and executable via JAR or direct compilation.

2. Data Pre-processing (SetUp)

- The program reads both Breast_train.csv and Breast_test.csv using custom CSV parsing logic to ensure robustness:
 - **Loading Process** (loadDataset method): Reads files line by line with Files.readAllLines. The normalizeCsvLines helper trims whitespace, skips empty lines and concatenates split rows (rows starting with commas). It then parses the header to extract feature names (age, menopause, tumor_size, etc.) and dynamically determines the feature count.
 - **Instance Parsing**: Each subsequent row is split by commas. The first column is the integer class label (0 or 1). Remaining columns are parsed as integer features using parseIntSafe (defaults to 0 for blanks/malformed values).
 - **Feature Statistics**: Tracks minValues and maxValues and (for decision trees) observedValues. Sets per feature. These enable bounded random constants in Arithmetic GP and realistic threshold sampling from observed values in Decision Tree GP. This optimisation avoids invalid splits and improves initialisation quality.
- No external libraries are used. Data is stored in a serializable Dataset object containing Instance lists for efficient evaluation.

3. Model Specifications

3.1 Genetic Programming Parameters

- All design decisions (dd) are explicitly defined as constants in both programs for clarity and reproducibility:

Parameter	Value	Implementation Notes
Population size	200	Fixed
Initial tree generation	Ramped half-and-half	Alternates full mode, random depths
Initial tree depth	4	INITIAL_TREE_DEPTH
Max. offspring depth	6	MAX_OFFSPRING_DEPTH; enforces via trimToDepth
Selection method	Tournament selection	tournamentSelect
Tournament size	5	TOURNAMENT_SIZE
Function set (Arithmetic)	+, -, *, /(protected)	Operator enum
Function set (DT)	Feature thresholds and leaves	Binary splits
Crossover rate	90%	CROSSOVER_RATE
Mutation rate	10%	MUTATION_RATE, point mutation
Mutation type	Point	Operator/feature/threshold/leaf flip
Mutation offspring depth	<= 6	Post-mutation trimming
Fitness function	Accuracy	Direct classification accuracy
Elitism	1	Best individual preserved
Maximum generations	100	Fixed

Both models use subtree crossover and depth-bounding mutation/trimming.

3.2 Arithmetic Classifier (Symbolic GP) - ArithmeticGP.java

- **Representation:** Trees with OperatorNode (binary arithmetic), FeatureNode and ConstantNode.
- **Evaluation:** Computes a raw score, then applies sigmoid for probability: class prediction ($\geq 0.5 = 1$).
- **Key Features:**
 - Protected division (safeDivide).
 - Constants sampled from feature range.
 - Pretty-printing of full expressions (e.g ((feature1 * 2.5) + feature3))
- **Evolution:** Standard GP with the parameters above.

3.3 Decision Tree Classifier (Logical GP) - DecisionTreeGP.java

- **Representation:** Node interface with DecisionNode (feature $\leq$ threshold splits) and LeafNode (0 or 1 prediction).
- **Evaluation:** Direct integer prediction by traversing the tree.
- **Key Features:**
 - Thresholds sampled only from observed feature values (more realistic than arbitrary splits).
 - majorityPrediction for depth trimming.
 - Human readable pretty-printing with indentation (if-then-else structure).
- **Evolution:** Identical GP framework but adapted to tree nodes.

Both support serialization (saveModel / loadModel) for training: testing workflow.

4. Experimental Setup & Instructions

4.1 README Instructions (README Summary)

- Execution requires no external dependencies and runs by executing a jar.
- The program interactively prompts for:
 - Mode (train/test)
 - Seed (for train)
- Training prints best individual + metrics per generation, saves .model file and outputs final train/test results.
- Testing loads the model and reports test metrics and confusion matrix.
- Full README included in submission.

4.2 Execution Methodology

- 30 independent runs per model with unique seeds.
- Best run (by test performance) selected for demonstration and reporting.
- Seed for best Arithmetic run: 26
- Seed for best Decision Tree run: 19

5. Results and Evaluation

5.1 Training Demonstration

The GP successfully evolves improving individuals over 100 generations. The Decision Tree produces compact, human readable trees that effectively use key features such as age, inv_nodes, breast_quad and irradiat.

5.2 Classification Performance

Comparison between best performance of Arithmetic GP and Decision Tree GP using seeds 26 and 19 respectively:

Algorithm	Training (%)	Test (%)	F-measure	Runtimes (s)
Arithmetic GP	84.70	63.95	0.6076	0.949
Decision Tree GP	83.61	65.12	0.6053	0.450

Best Arithmetic Expression:

```
(((( (-3.714 + tumor_size) / irradiat) / (irradiat / (irradiat / -5.188))) + (-7.984 + ((deg_malign * deg_malign) - (node_caps * -7.815)))) + (-8.221 - ((irradiat / (irradiat / -7.984)) / inv_nodes)))
```

Best Decision Tree:

```
if breast_quad <= 2
  if deg_malign <= 2
    if node_caps <= 1
      The prediction is: 0
    else
      if age <= 1
        The prediction is: 1
      else
        The prediction is: 1
    else
      if node_caps <= 0
        if irradiat <= 0
          The prediction is: 0
        else
          The prediction is: 1
      else
        The prediction is: 1
    else
```

```

if irradiat <= 0
  if node_caps <= 0
    if deg_malig <= 2
      if node_caps <= 1
        The prediction is: 0
      else
        The prediction is: 1
    else
      if node_caps <= 0
        The prediction is: 1
      else
        The prediction is: 1
  else
    if age <= 4
      The prediction is: 0
    else
      if age <= 4
        The prediction is: 1
      else
        The prediction is: 0
  else
    The prediction is: 1

```

6. Statistical Significance

To determine whether the observed performance difference between the Arithmetic GP and Decision Tree GP is statistically meaningful (rather than due to random variation inherent in evolutionary algorithms), a Wilcoxon signed-rank test was performed. This is a type of test is well suited for Genetic Programming since:

- GP performance distributions are often non-normal.
- We are comparing paired results (same seeds or comparable random conditions across 30 independent runs).
- It does not assume normality or equal variance.

6.1 Experimental Design for Statistical Test

- 30 independent runs were executed for each algorithm.
- Each run used a unique random seed (seeds 1 through 30 for systematic comparison).
- All other parameters (population size = 200, generations = 100, etc.) were kept identical.
- Test accuracy and F-measure on the unseen test set were recorded for each run.

6.2 Summary Statistics from 30 Runs

Metric	Arithmetic GP	Decision Tree GP
Mean Test Accuracy	50.74%	55.43%
Best Test Accuracy	63.95%	65.12%
Standard Deviation	6.27%	7.51%

6.3 Wilcoxon Signed-Rank Test Results

Null Hypothesis (H_0): There is no difference in median test accuracy between Arithmetic GP and Decision Tree GP.

Alternative Hypothesis (H_A): There is a difference in median test accuracy.

Metric	Test Statistic	p-value	Significance ($\alpha = 0.05$)	Conclusion
Test Accuracy	72.5	0.0051	Significant	Reject H_0

Interpretation:

The p-value of 0.0051 is well below the 0.05 significance level. Therefore, we reject the null hypothesis and conclude that there is a statistically significant difference in performance between the two models. The Decision Tree GP demonstrates superior generalisation ability on the unseen test set.